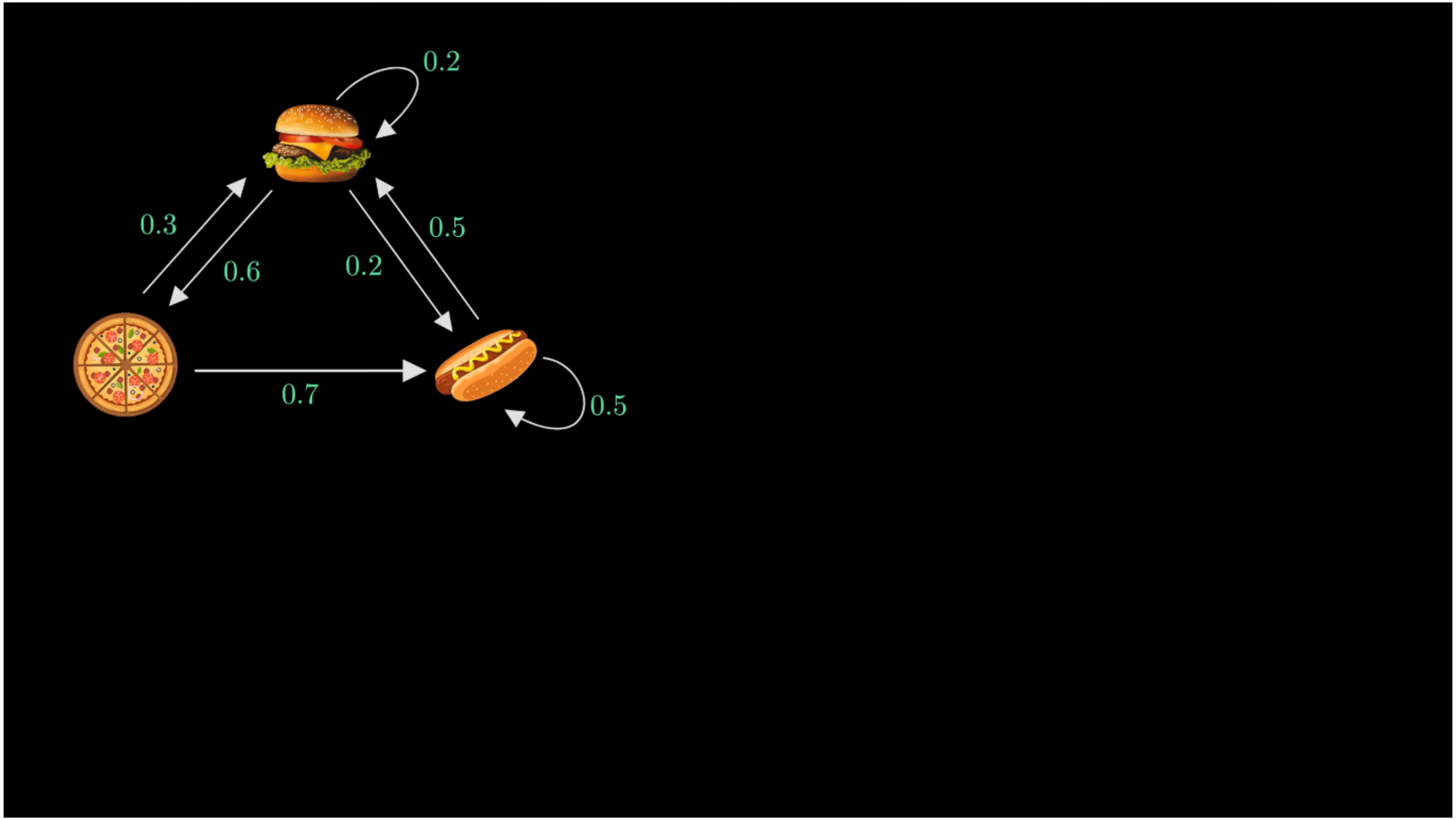
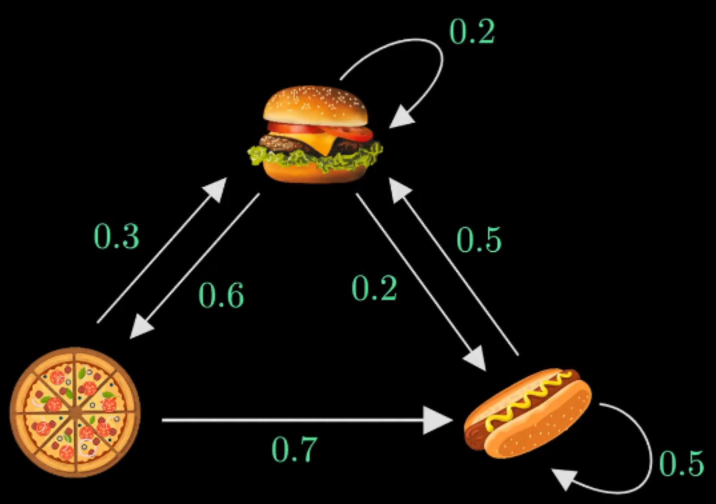
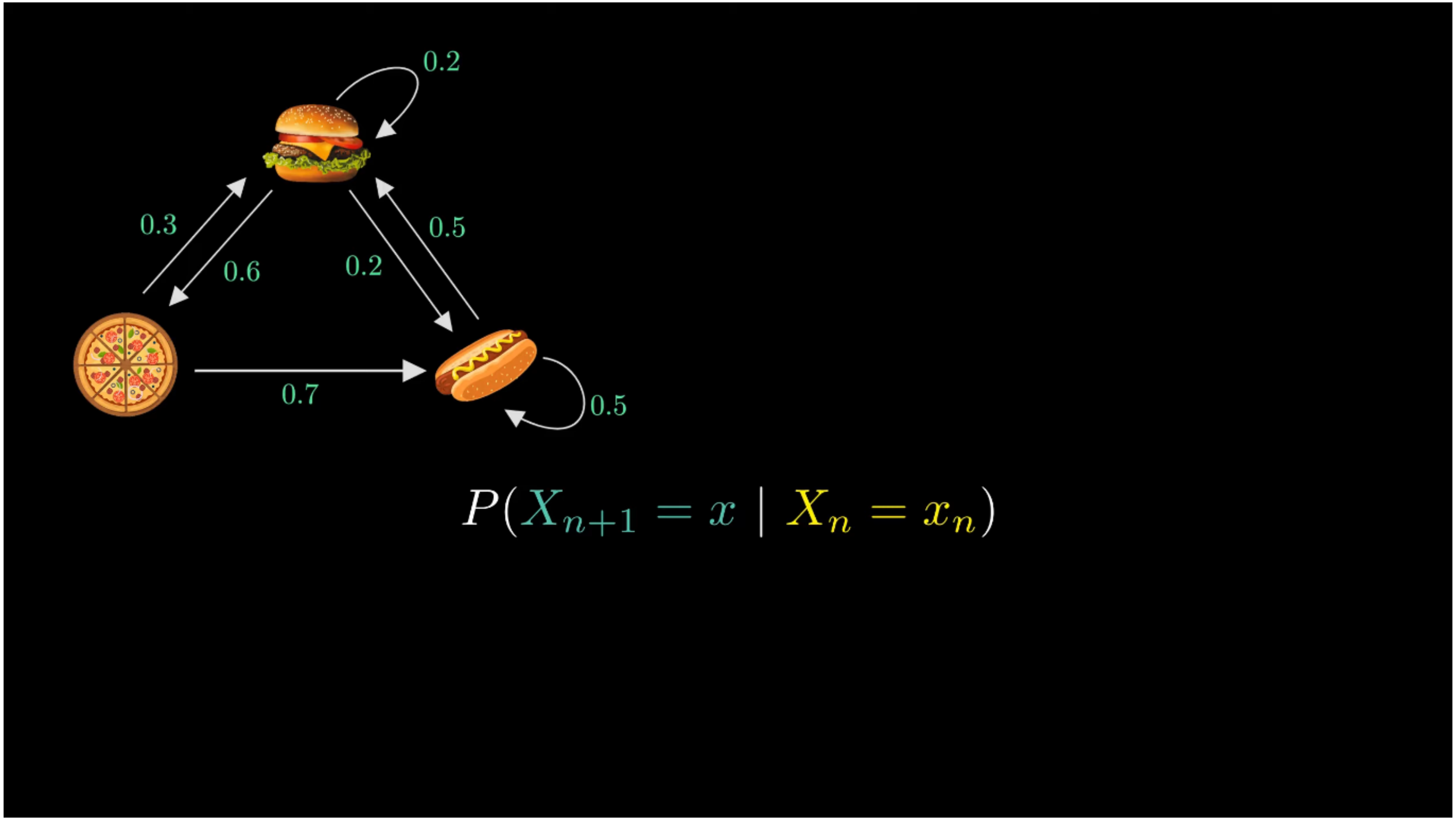
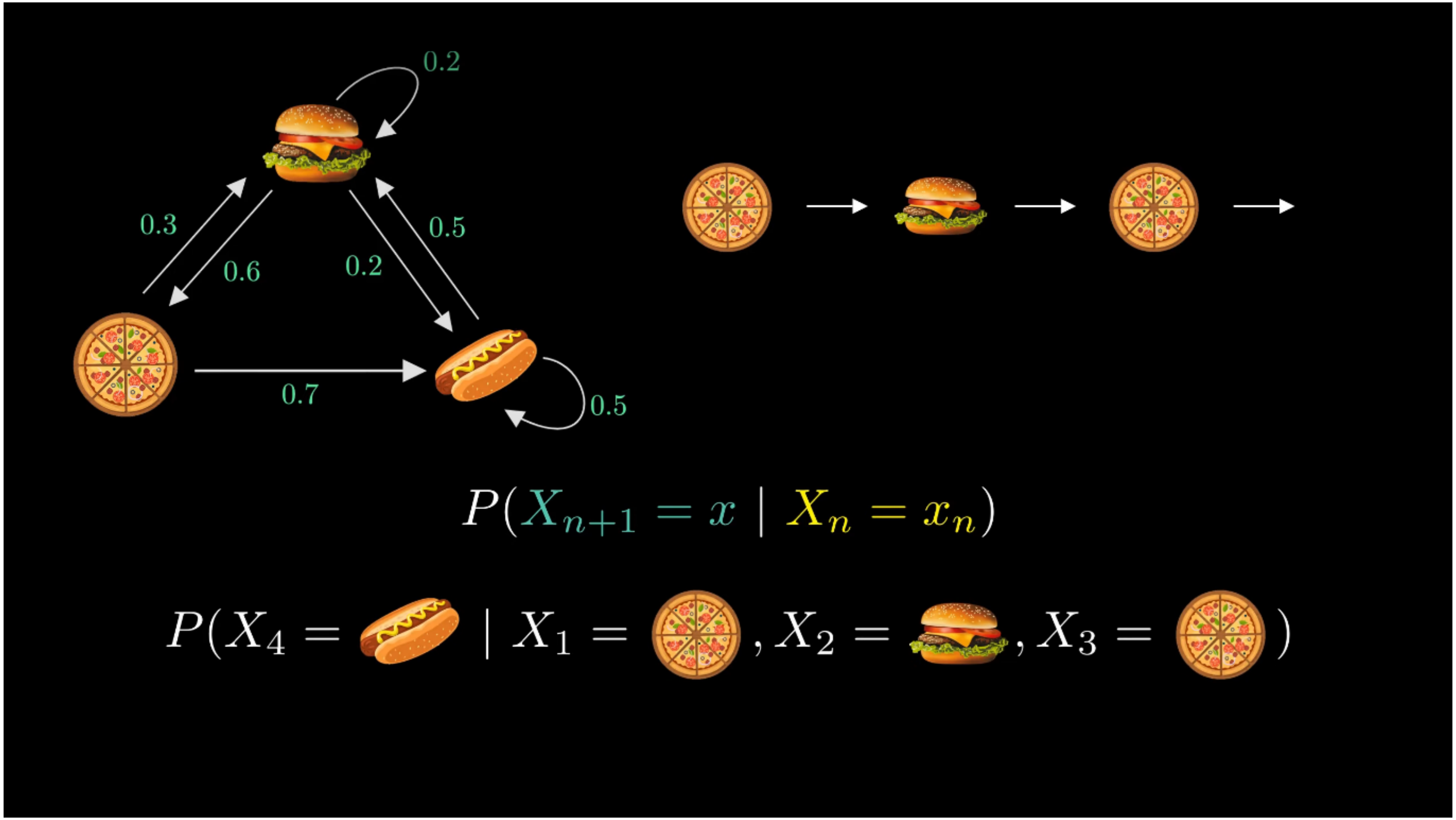
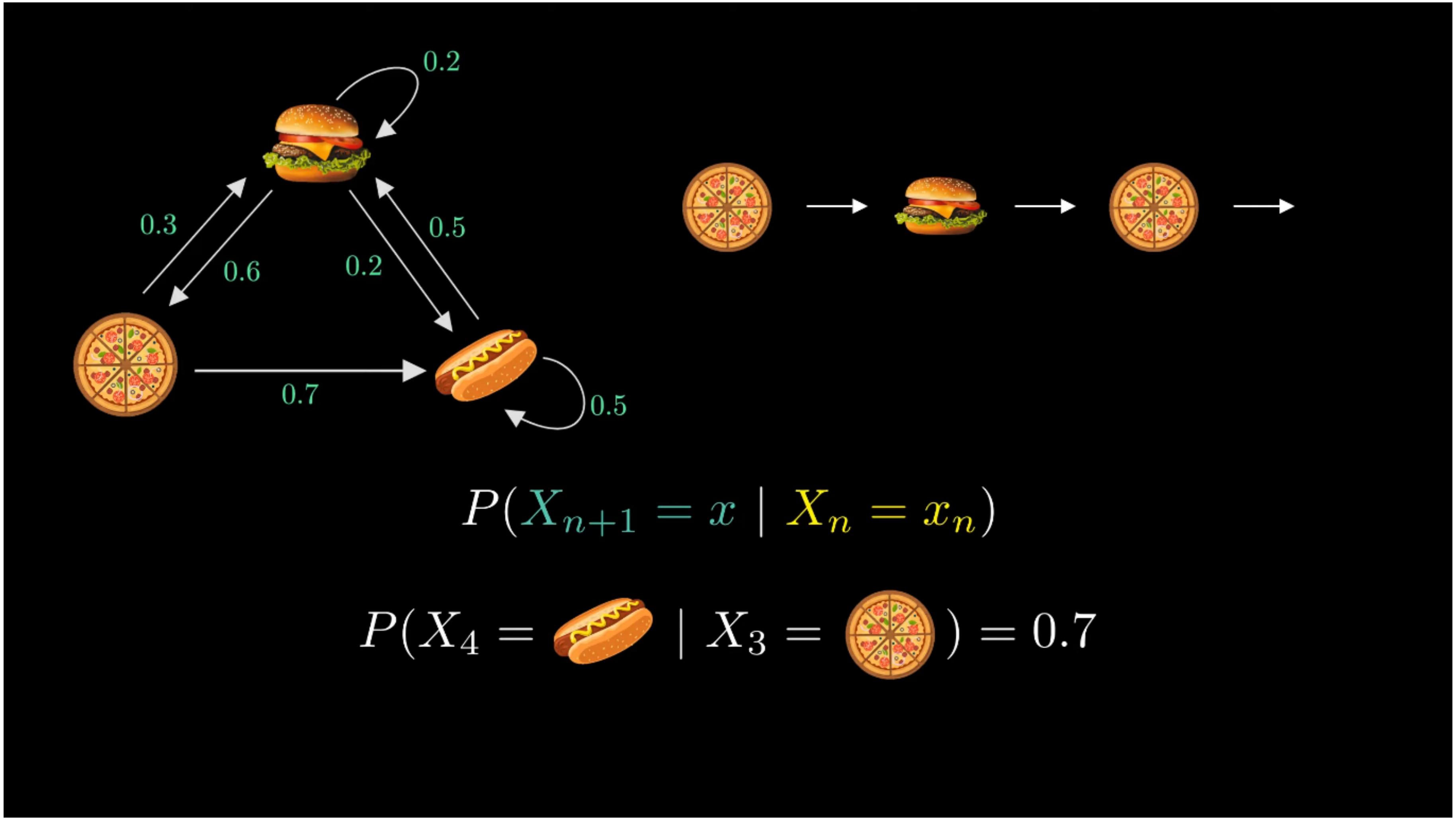

0.2
0.3
0.6
0.2
0.5
0.7
0.5

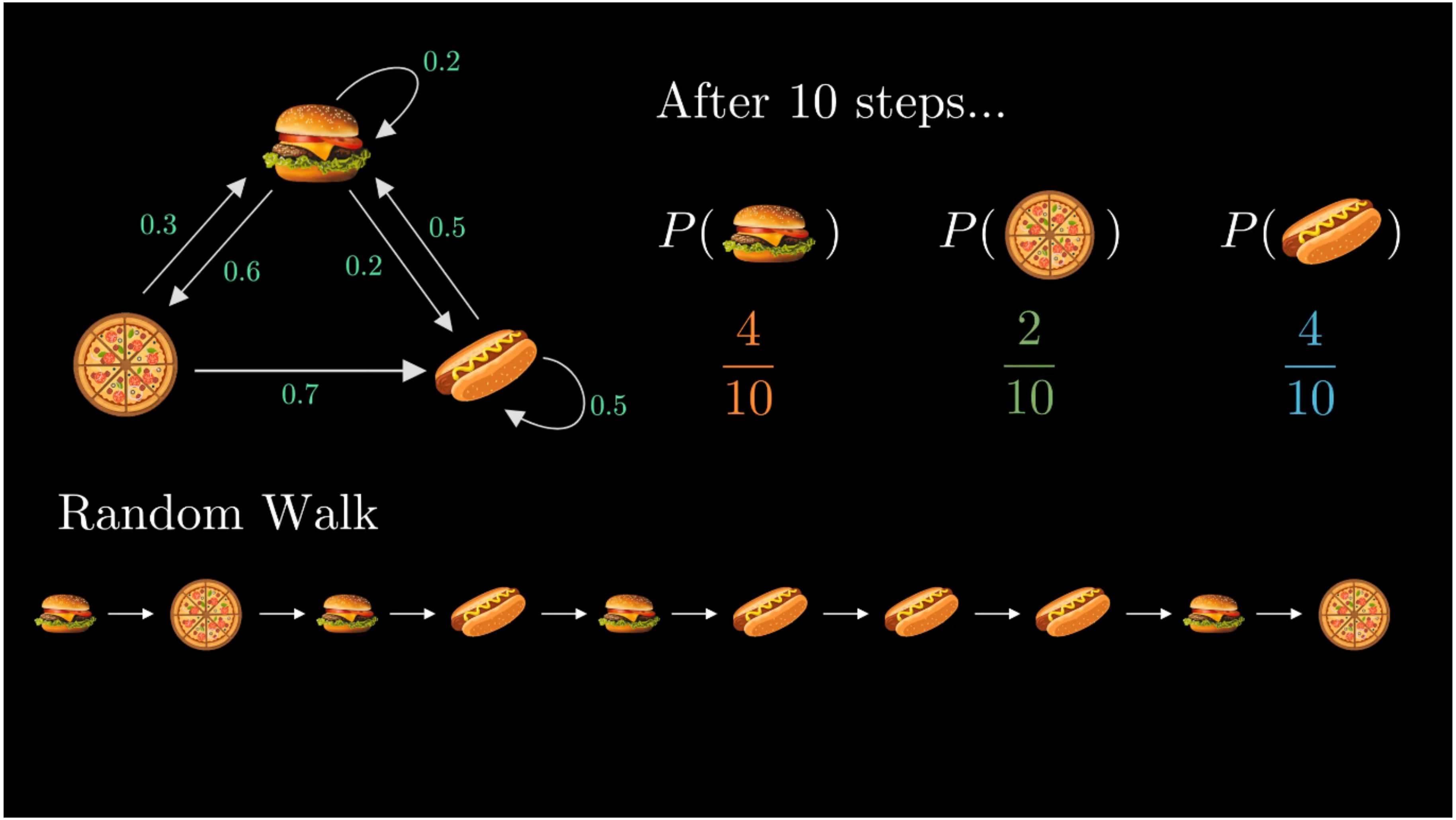


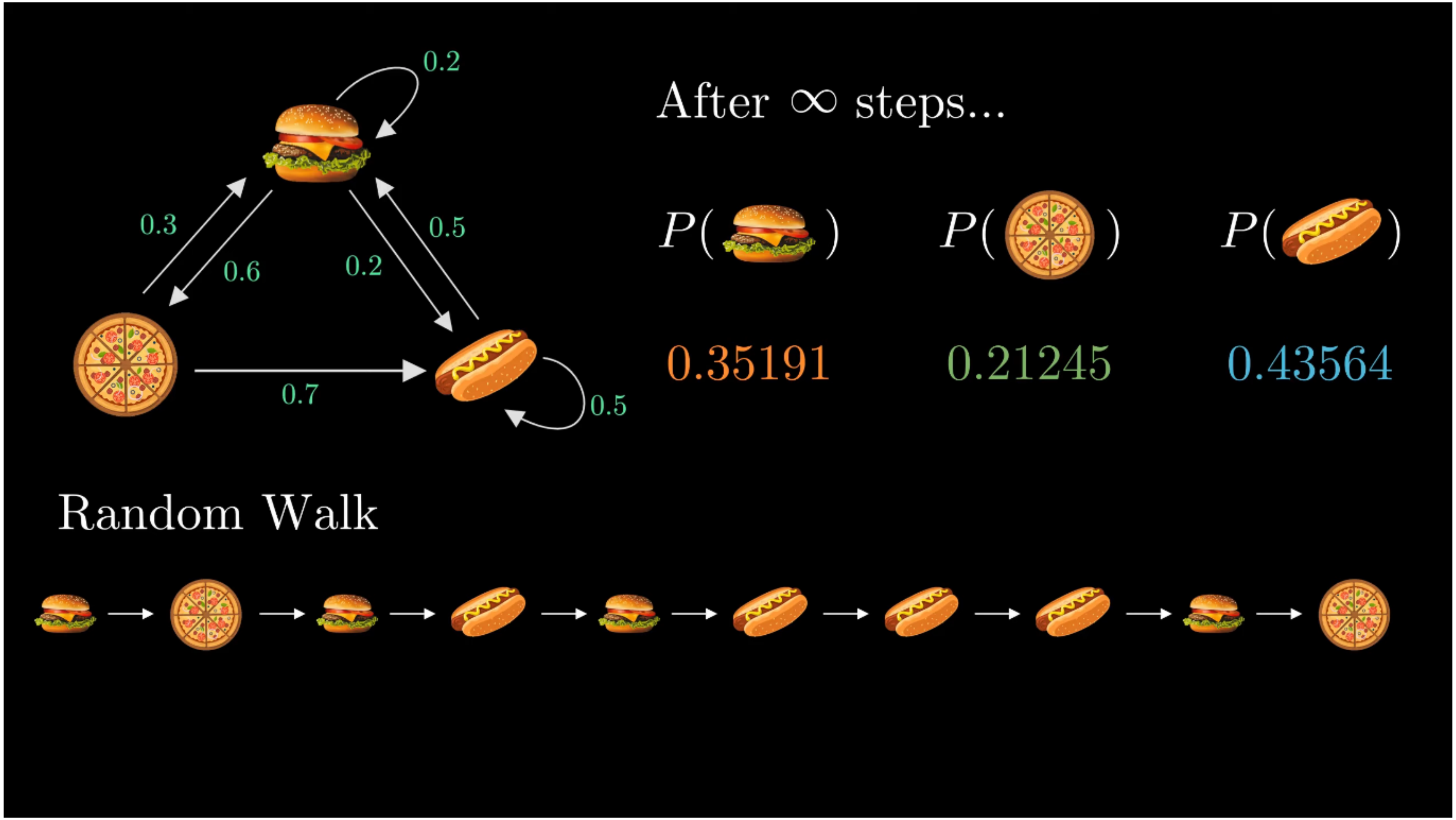
$$P(X_{n+1} = x \mid X_1 = x_1, X_2 = x_2, \ldots, X_n = x_n)$$

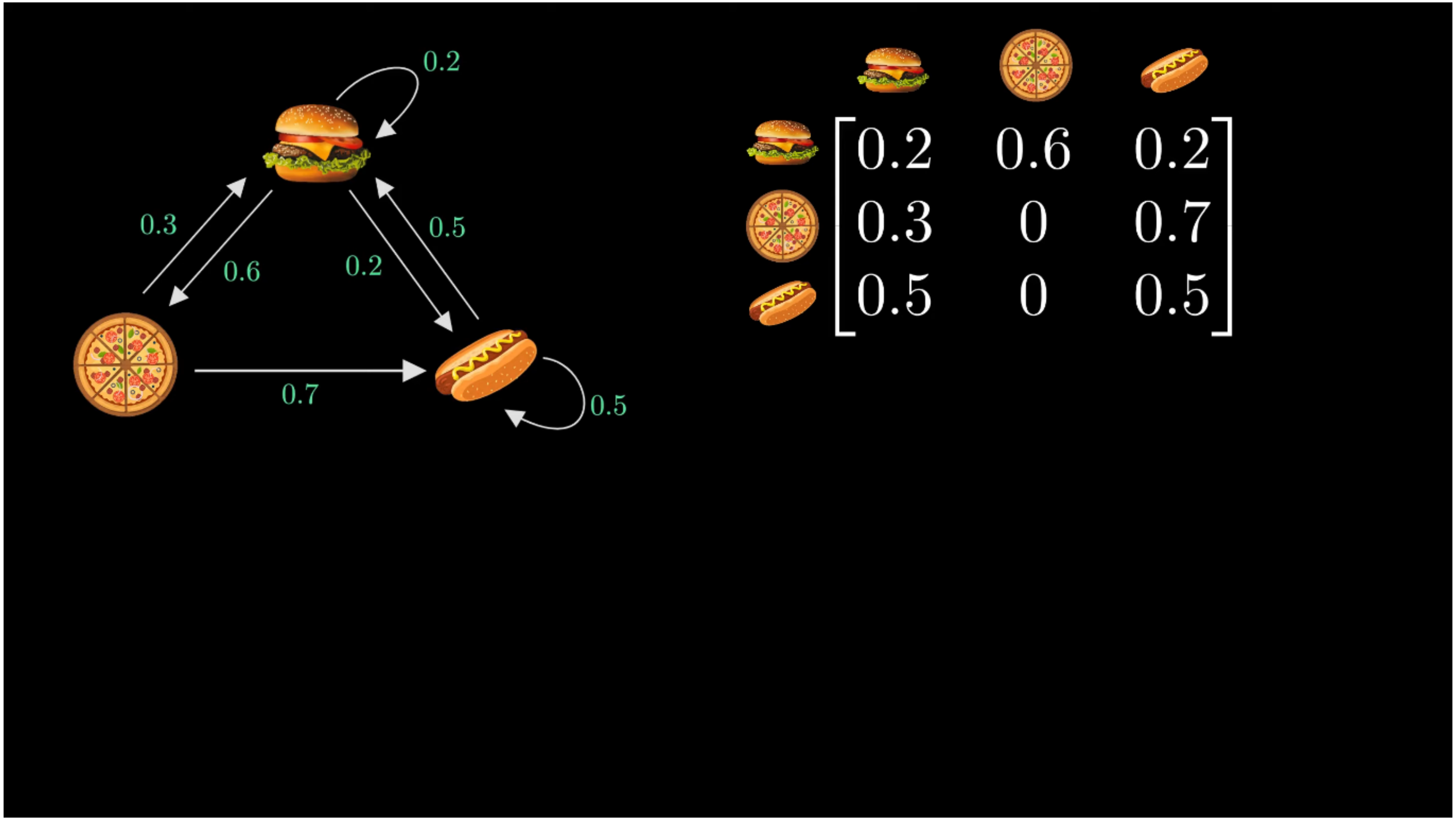

0.2
0.3
0.6
0.2
0.5
0.7
0.5
$P(X_{n+1} = x \mid X_n = x_n)$

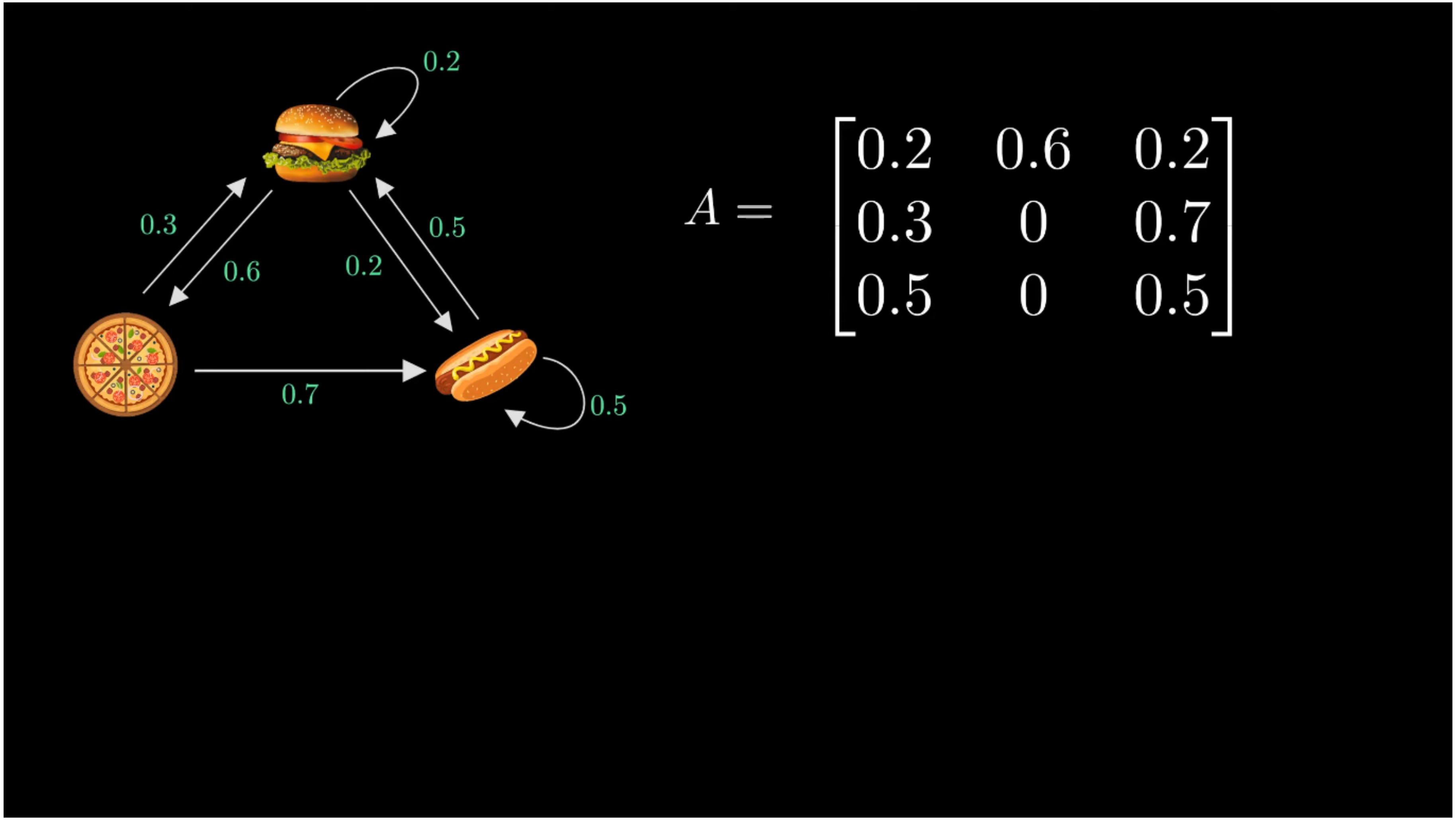

0.2
0.3
0.6
0.2
0.5
0.7
0.5
$P(X_{n+1} = x \mid X_n = x_n)$
$P(X_4 =$
$\mid X_1 =$
$, X_2 =$
$, X_3 =$
$)$

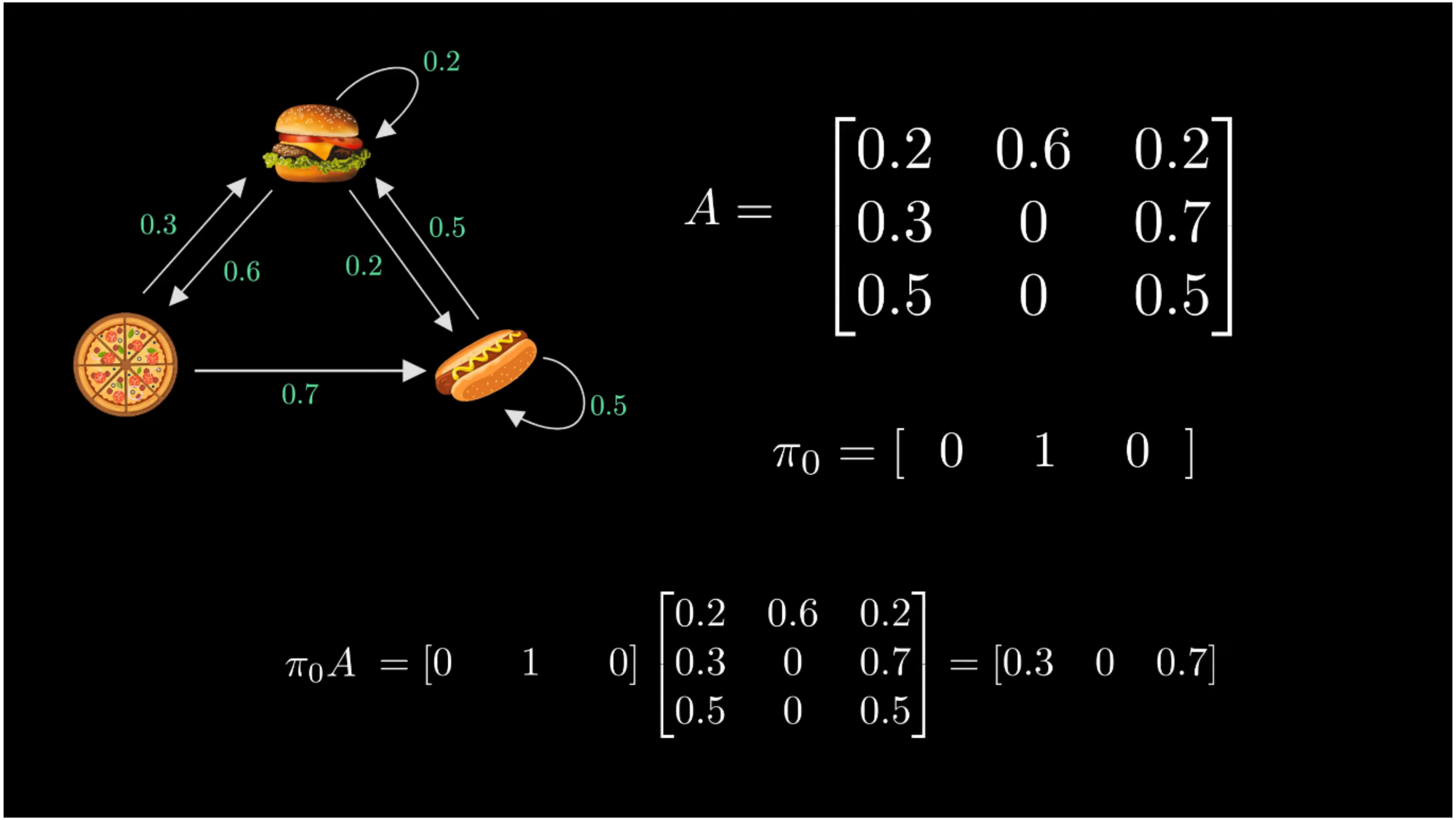

0.2
0.3
0.6
0.2
0.5
0.7
0.5
$P(X_{n+1} = x \mid X_n = x_n)$
$P(X_4 = \quad \mid X_3 = \quad) = 0.7$

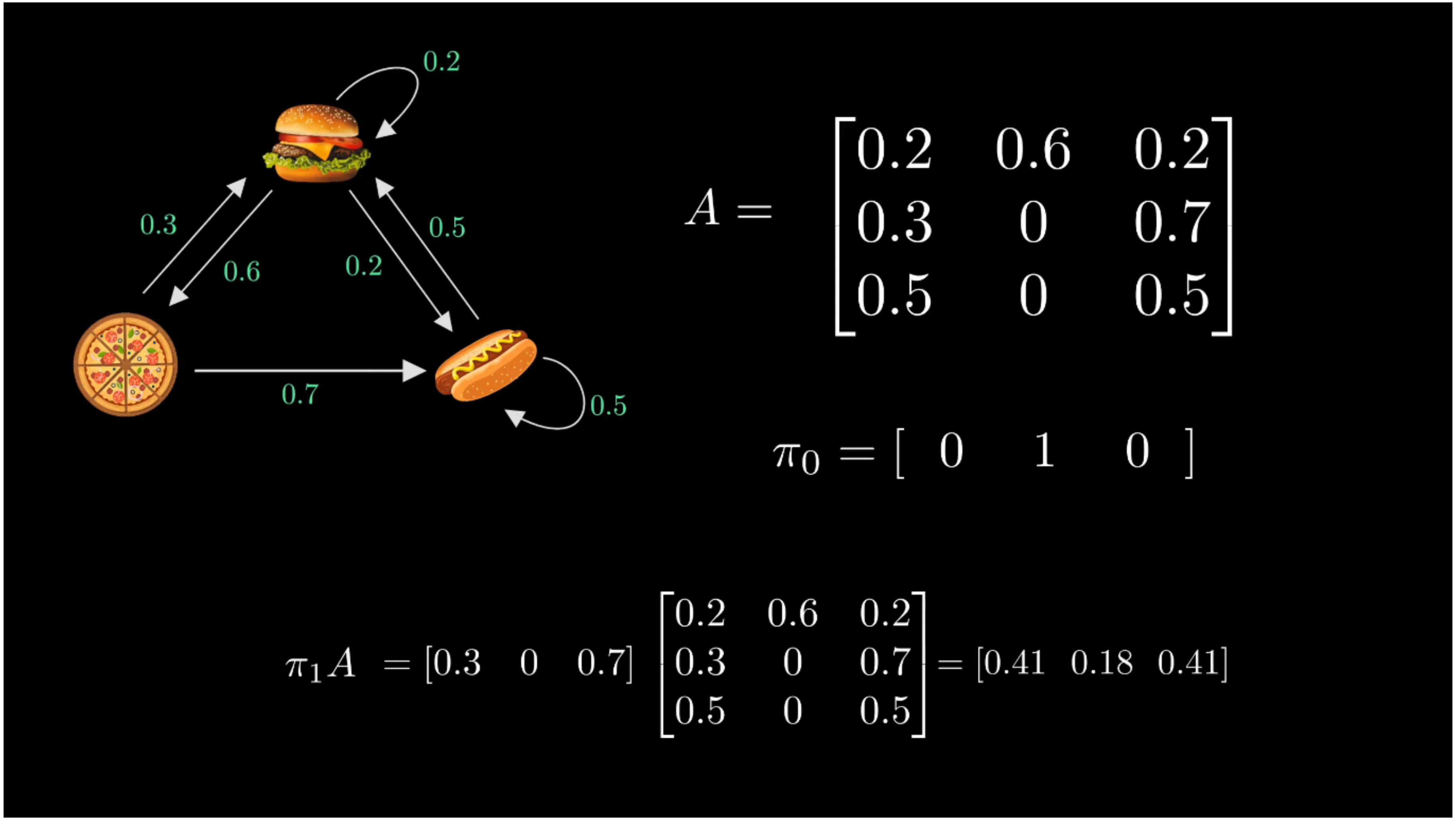

0.2
After 10 steps...
0.3
0.5
0.6
0.2
0.7
0.5
$P(\,)$ $\frac{4}{10}$
$P(\,)$ $\frac{2}{10}$
$P(\,)$ $\frac{4}{10}$
Random Walk

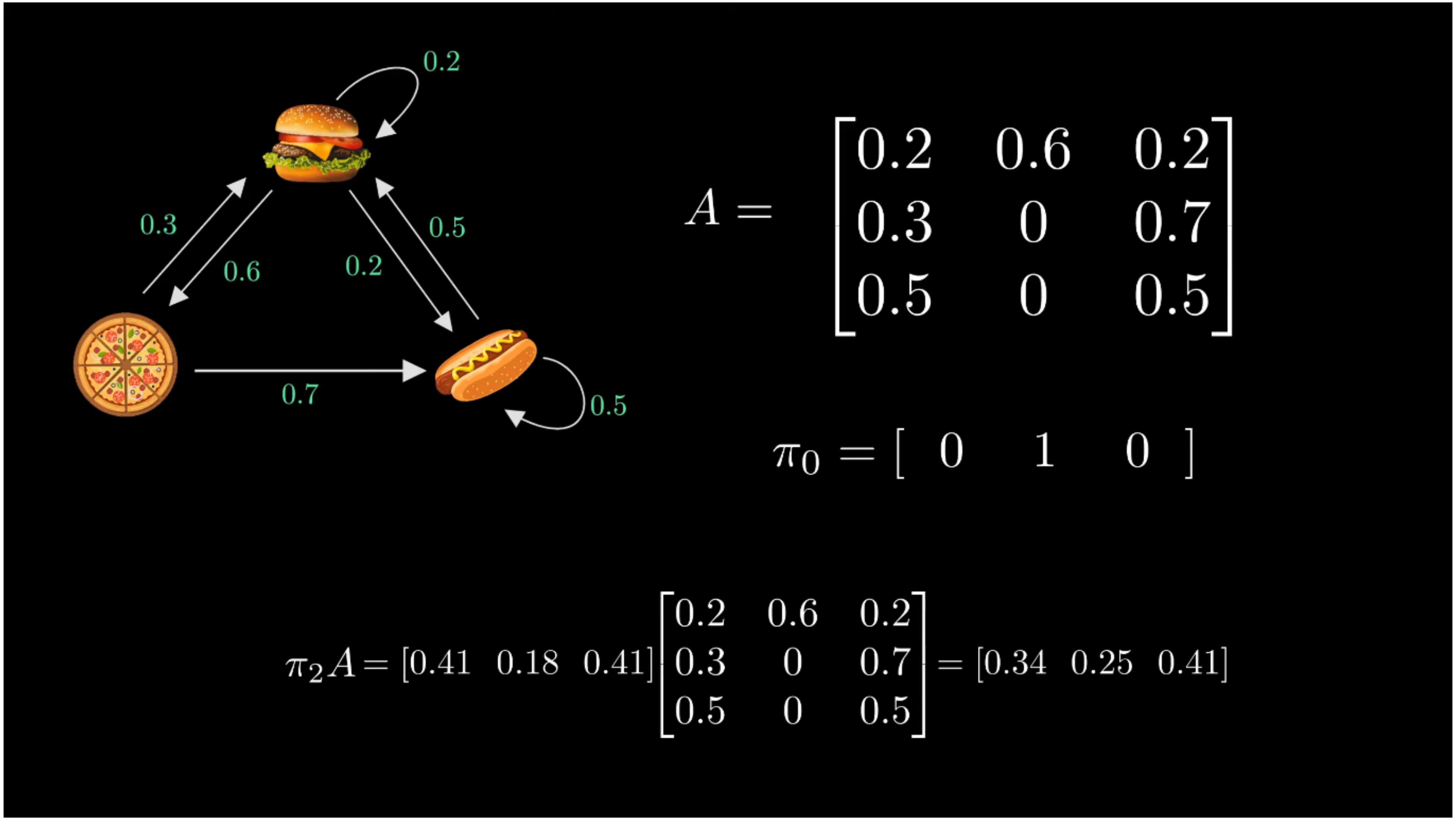

0.2
0.3
0.6
0.2
0.5
0.7
0.5
After ∞ steps...
P( )
P( )
P( )
0.35191
0.21245
0.43564
Random Walk

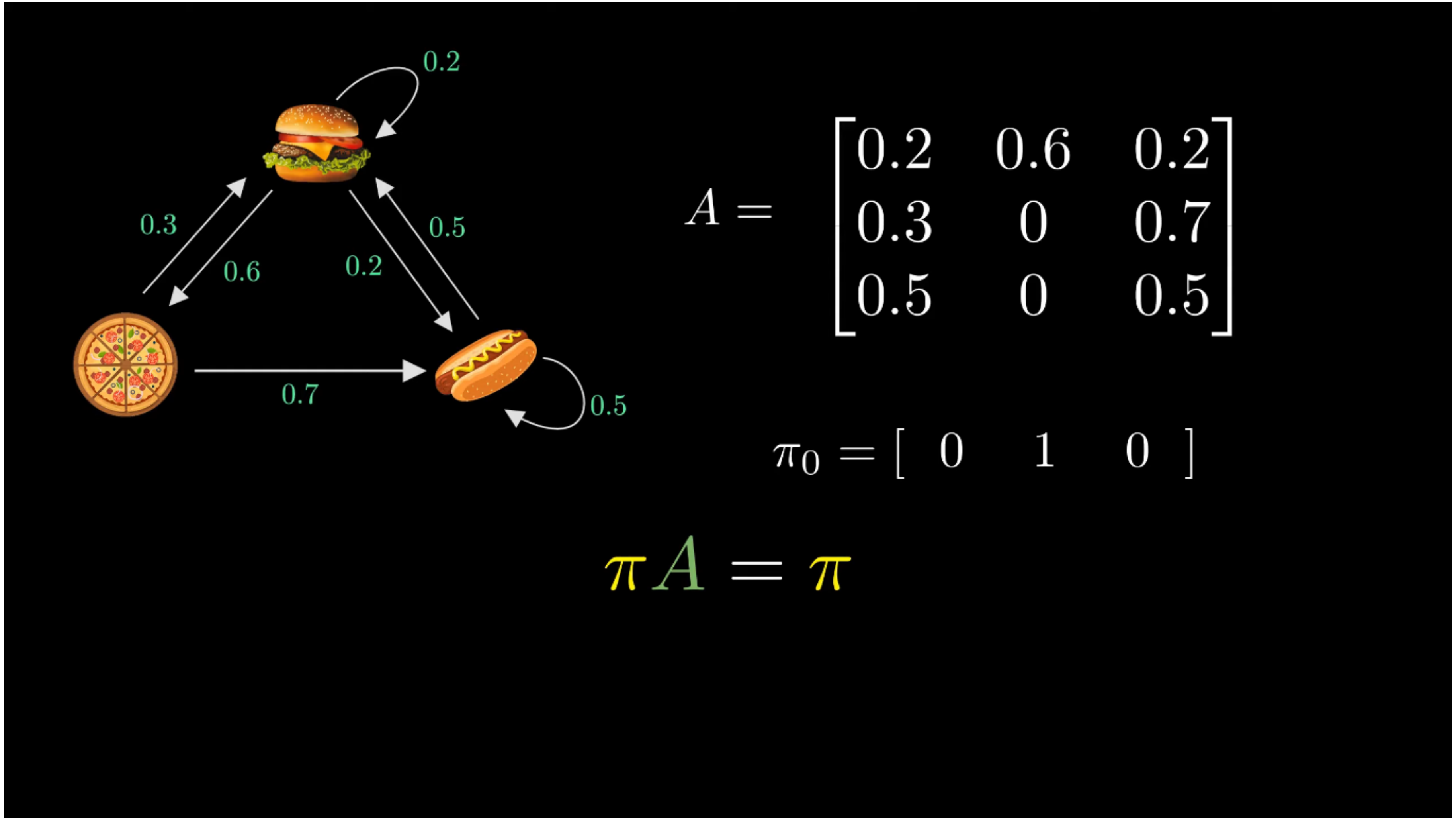

0.2
0.3
0.6
0.2
0.5
0.7
0.5
0.2 0.6 0.2
0.3 0 0.7
0.5 0 0.5

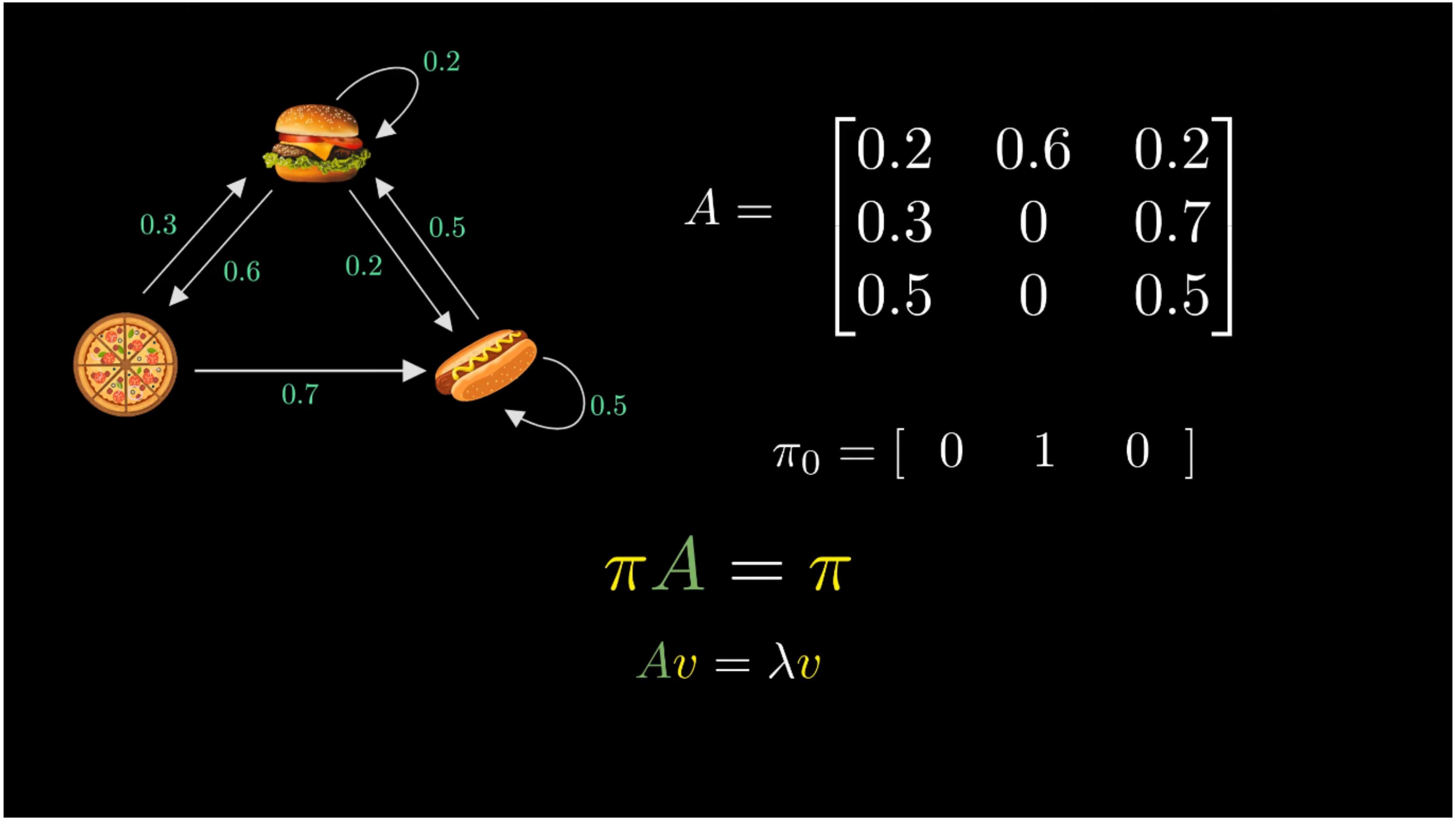

0.2
0.3
0.6
0.2
0.5
0.7
0.5
$A = \begin{bmatrix} 0.2 & 0.6 & 0.2 \\ 0.3 & 0 & 0.7 \\ 0.5 & 0 & 0.5 \end{bmatrix}$

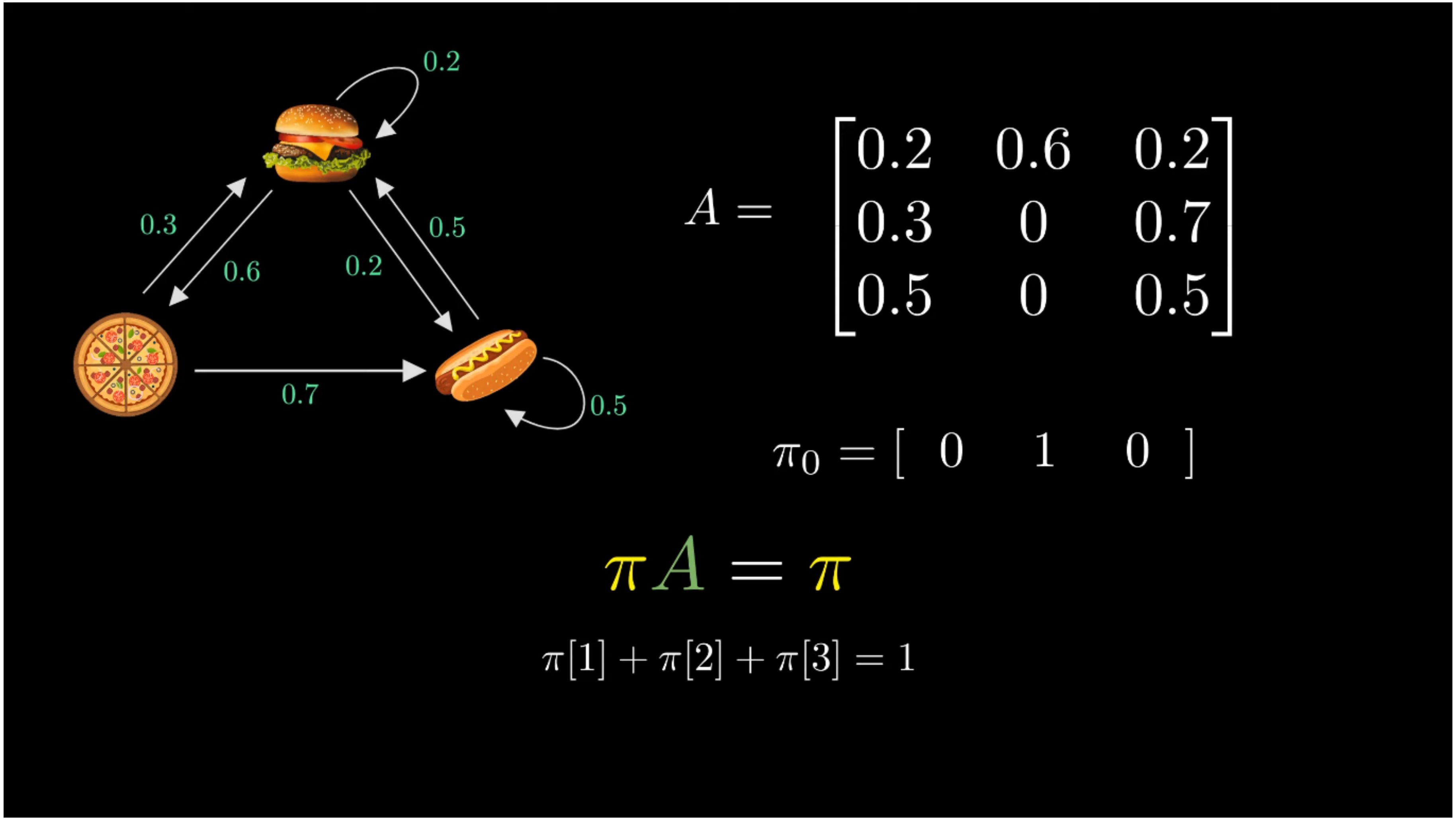

0.2
0.3
0.6
0.2
0.5
0.7
0.5
$A = \begin{bmatrix} 0.2 & 0.6 & 0.2 \\ 0.3 & 0 & 0.7 \\ 0.5 & 0 & 0.5 \end{bmatrix}$
$\pi_0 = [\ 0 \quad 1 \quad 0\ ]$
$\pi_0 A = [0 \quad 1 \quad 0] \begin{bmatrix} 0.2 & 0.6 & 0.2 \\ 0.3 & 0 & 0.7 \\ 0.5 & 0 & 0.5 \end{bmatrix} = [0.3 \quad 0 \quad 0.7]$

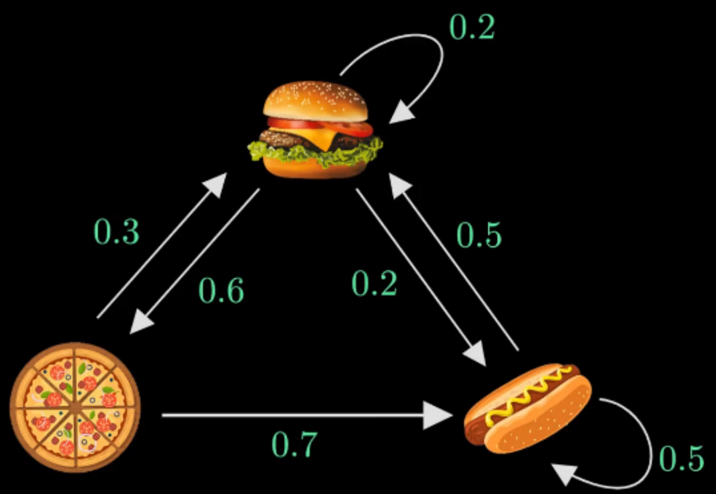

0.2
0.3
0.6
0.2
0.5
0.7
0.5
$A = \begin{bmatrix} 0.2 & 0.6 & 0.2 \\ 0.3 & 0 & 0.7 \\ 0.5 & 0 & 0.5 \end{bmatrix}$
$\pi_0 = [\ 0 \quad 1 \quad 0\ ]$
$\pi_1 A = [0.3 \quad 0 \quad 0.7] \begin{bmatrix} 0.2 & 0.6 & 0.2 \\ 0.3 & 0 & 0.7 \\ 0.5 & 0 & 0.5 \end{bmatrix} = [0.41 \quad 0.18 \quad 0.41]$


0.2
0.3
0.6
0.2
0.5
0.7
0.5
$A = \begin{bmatrix} 0.2 & 0.6 & 0.2 \\ 0.3 & 0 & 0.7 \\ 0.5 & 0 & 0.5 \end{bmatrix}$
$\pi_0 = [\ 0 \quad 1 \quad 0\ ]$
$\pi_2 A = [0.41 \quad 0.18 \quad 0.41] \begin{bmatrix} 0.2 & 0.6 & 0.2 \\ 0.3 & 0 & 0.7 \\ 0.5 & 0 & 0.5 \end{bmatrix} = [0.34 \quad 0.25 \quad 0.41]$


0.2
0.3
0.6
0.2
0.5
0.7
0.5
$A = \begin{bmatrix} 0.2 & 0.6 & 0.2 \\ 0.3 & 0 & 0.7 \\ 0.5 & 0 & 0.5 \end{bmatrix}$
$\pi_0 = [\ 0 \quad 1 \quad 0\ ]$
$\pi A = \pi$


0.2
0.3
0.6
0.2
0.5
0.7
0.5
$A = \begin{bmatrix} 0.2 & 0.6 & 0.2 \\ 0.3 & 0 & 0.7 \\ 0.5 & 0 & 0.5 \end{bmatrix}$
$\pi_0 = [\ 0 \quad 1 \quad 0\ ]$
$\pi A = \pi$
$Av = \lambda v$

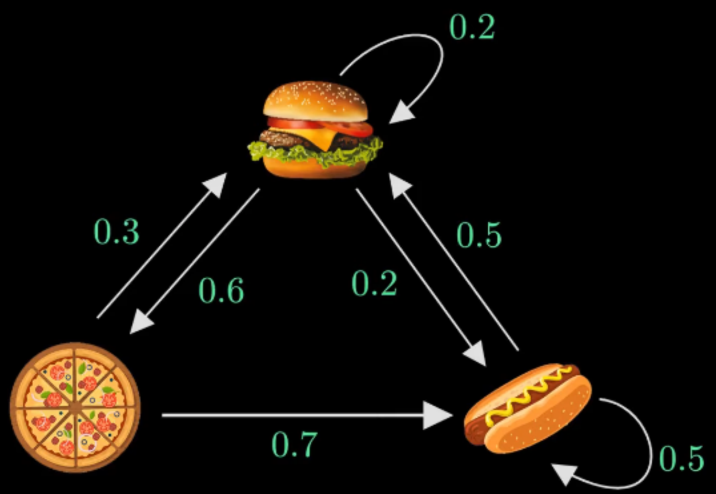

0.2
0.3
0.6
0.2
0.5
0.7
0.5
$A = \begin{bmatrix} 0.2 & 0.6 & 0.2 \\ 0.3 & 0 & 0.7 \\ 0.5 & 0 & 0.5 \end{bmatrix}$
$\pi_0 = [\ 0 \quad 1 \quad 0\ ]$
$\pi A = \pi$
$\pi[1] + \pi[2] + \pi[3] = 1$



$$A = \begin{bmatrix} 0.2 & 0.6 & 0.2 \\ 0.3 & 0 & 0.7 \\ 0.5 & 0 & 0.5 \end{bmatrix}$$

$$\pi_0 = \begin{bmatrix} 0 & 1 & 0 \end{bmatrix}$$

$$\pi = \begin{bmatrix} 0.35211 & 0.21127 & 0.43662 \end{bmatrix}$$



$$A = \begin{bmatrix} 0.2 & 0.6 & 0.2 \\ 0.3 & 0 & 0.7 \\ 0.5 & 0 & 0.5 \end{bmatrix}$$

$$\pi_0 = \begin{bmatrix} 0 & 1 & 0 \end{bmatrix}$$

$$\pi = \begin{bmatrix} 0.35211 & 0.21127 & 0.43662 \end{bmatrix}$$

$$0.35191 \quad 0.21245 \quad 0.43564$$